

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2141

Roanoke Rapids 69 kV
December 24, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2141. The Interconnection Customer proposes a 109.2 MW battery energy storage facility interconnecting at the Roanoke Rapids 69 kV substation in the North Carolina area. The request was received on July 22, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 617 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$2,400,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Roanoke Rapids Substation 138 kV/69 kV Transformer	Shaw Hills 138 kV/1 kV Transformer Bank	124.4	62	45.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Roanoke Rapids 138 kV/69 kV Transformer Bank	Replace transformer with higher-rated unit	\$4,551,000	53% (\$2,400,000)
	Total allocated to project		\$2,400,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.